SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I **N. HEMA SREE (192511100)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **N. HEMA SREE**

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

ANSWER:

<u>Problem Analysis</u>

For a 20-person startup, a reliable, scalable, and easy-to-manage network is required. The network should support workstations, printers, VoIP phones, laptops, and other devices while providing secure Internet connectivity. Faults should also be easy to identify without affecting the entire network.

<u>Solution Design</u>

A **star topology** is the most suitable network topology for this startup. In this design, all devices are connected individually to a **central managed switch**. A 24-port Layer 2/Layer 3 managed switch can be used, providing enough ports for the 20 employees and additional ports for future expansion.

The switch is connected to a **router/firewall**, which provides Internet access, DHCP, and Network Address Translation (NAT). A **Wi-Fi access point** is also connected to the switch to provide wireless connectivity for laptops and mobile devices.

<u>Network structure:</u>

Internet → Router/Firewall → Managed Switch →

- Workstations
- Printers
- VoIP Phones
- Wi-Fi Access Point
- Laptops/Mobile Devices

<u>Technical Implementation</u>

Each workstation and network device is connected to the central switch using **Cat6 UTP cables**. Cat6 supports Gigabit Ethernet and is suitable for typical office distances up to 100 meters.

A **24-port managed switch** is selected because it provides sufficient ports for the current 20-person workforce while allowing additional devices to be connected later. VLANs can also be configured to separate departments or services such as employees, guests, and VoIP devices.

The router/firewall provides:

- Internet connectivity
- DHCP address allocation
- NAT
- Basic firewall protection

The Wi-Fi access point provides wireless connectivity to mobile devices and laptops.

Use of Tools and Technologies

The following networking technologies are used:

- **Star topology** – provides centralized network management.
- **Managed Layer 2/Layer 3 switch** – connects and manages network devices.
- **Cat6 UTP cables** – provide high-speed wired connectivity.
- **Router/Firewall** – provides Internet access, NAT, DHCP, and security.
- **Wi-Fi Access Point** – provides wireless connectivity.
- **VLANs** – can be used for network segmentation and improved security.

Problem-Solving Approach

The design provides effective **fault isolation**. If one workstation's cable or network interface fails, only that particular device is disconnected. The remaining 19 employees can continue using the network.

Centralized management also makes troubleshooting easier because the administrator can monitor connections through the managed switch. Additional devices can be added without redesigning the complete network.

Innovation and Scalability

The network is designed for future growth. Although the startup currently has 20 employees, the 24-port switch provides additional ports for expansion. VLANs, additional access points, and higher-capacity switches can be introduced as the company grows.

A managed network also allows administrators to monitor traffic, control access, and improve network security.

Testing and Validation

The network can be tested using:

1. **Ping test** – to verify connectivity between devices.
2. **IP configuration check** – to verify correct IP addresses and gateway settings.
3. **Internet connectivity test** – to confirm communication through the router.
4. **Cable testing** – to identify faulty Cat6 cables.
5. **Wi-Fi testing** – to verify wireless coverage and connectivity.

6. **Device failure test** – disconnect one workstation and verify that other devices remain connected.

Documentation and Presentation

The network topology should be documented using a network diagram showing the Internet, router/firewall, managed switch, workstations, printers, VoIP phones, and Wi-Fi access point. IP addressing, VLAN configuration, cable connections, and device details should also be recorded for future maintenance.

Teamwork and Collaboration

The implementation can be divided among team members. One member can configure the router and firewall, another can configure the managed switch and VLANs, another can set up the Wi-Fi access point, and another can perform connectivity testing and documentation.

Conclusion

A **star topology** is an ideal choice for a 20-person startup because it provides centralized management, easy troubleshooting, good performance, fault isolation, and scalability. Using a managed switch, Cat6 UTP cables, router/firewall, and Wi-Fi access point creates a reliable and secure network that can support the startup's current requirements and future growth.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

ANSWER:

Problem Analysis

A banking organization requires a network with **near-zero downtime, high reliability, security, and continuous availability** because financial transactions and customer services must operate without interruption. A single network link or device failure should not stop critical banking operations.

Solution Design

A **full mesh or partial mesh topology** is suitable for a banking network. In a full mesh, every important network device has a direct connection to every other device. However, a **partial mesh topology** is more practical and cost-effective for large banking networks.

The core layer can connect multiple core switches, routers, firewalls, and critical servers using redundant connections. Distribution and access switches can have **dual uplinks** to two different core switches.

Network structure:

Internet → Dual Routers → Core Switches/Firewalls → Distribution Switches → Access Switches → Servers & User Devices

This design provides multiple paths for data transmission and prevents a single point of failure.

Technical Implementation

Two enterprise-grade routers can be deployed to provide redundant Internet connectivity. **BGP (Border Gateway Protocol)** can be used for redundant connections to different Internet service providers.

At the internal network level, **OSPF (Open Shortest Path First)** can dynamically select alternative paths when a network link fails.

High-speed **fiber optic cables** should be used between core network devices because they provide high bandwidth, low latency, and immunity to electromagnetic interference.

- **Single-mode fiber:** Suitable for long-distance connections between buildings.
- **Multi-mode OM4 fiber:** Suitable for high-speed connections within data centers.
- **Redundant power supplies:** Keep critical devices operating if one power supply fails.
- **UPS systems:** Provide backup power during electrical failures.
- **RAID storage:** Protects critical data against individual disk failures.

Use of Tools and Technologies

Important technologies used in the banking network include:

- **Partial/Full Mesh Topology** – provides multiple communication paths.
- **BGP** – provides redundant Internet routing.
- **OSPF** – provides dynamic internal routing.
- **Fiber Optic Cables** – provide high-speed and reliable communication.
- **Redundant Routers** – prevent Internet connectivity failure.
- **Redundant Firewalls and Switches** – improve availability and security.
- **UPS** – provides backup power.
- **RAID** – improves server storage reliability.

Problem-Solving Approach

The main problem is preventing network downtime caused by device or link failures. The mesh topology solves this problem by providing **multiple alternative paths**.

For example, if one link between two core switches fails, routing protocols can identify the failure and redirect traffic through another available path. Similarly, if one core switch becomes unavailable, the redundant core switch can continue handling network traffic.

This eliminates major single points of failure and improves the reliability of banking services.

Innovation and Scalability

The network can be designed with redundant components so that additional routers, switches, firewalls, and servers can be integrated without affecting existing services.

Network monitoring systems can also be used to continuously monitor bandwidth, device health, link status, and failures. Automatic failover mechanisms can reduce service interruption and improve overall network availability.

Testing and Validation

The network should be tested using the following methods:

1. **Link failure test** – disconnect one network link and verify that traffic uses an alternative path.
2. **Router failure test** – shut down one router and verify that the backup router provides connectivity.
3. **Switch failure test** – verify that redundant core switches continue network operation.
4. **Ping and traceroute tests** – verify connectivity and routing paths.
5. **BGP/OSPF verification** – confirm that routing protocols select alternative paths correctly.
6. **Power failure test** – verify that UPS systems maintain operation during power interruptions.
7. **Performance testing** – measure latency, bandwidth, and packet loss.

Documentation and Presentation

The complete network should be documented using a topology diagram showing routers, firewalls, core switches, distribution switches, servers, Internet connections, and redundant links.

The documentation should also contain IP addressing, routing configuration, fiber connections, redundancy mechanisms, backup procedures, and testing results. Proper documentation helps network administrators troubleshoot and maintain the banking infrastructure.

Teamwork and Collaboration

The implementation can be divided among network administrators and engineers. One team can configure routers and BGP, another can configure switches and OSPF, another can manage firewalls and security, and another can perform testing, monitoring, and documentation.

Conclusion

A **partial mesh topology with redundant core devices and links** is an excellent choice for a banking network because it provides high availability, fault tolerance, scalability, and fast recovery from failures. Technologies such as **BGP, OSPF, fiber optics, redundant routers, UPS systems, and RAID** further improve reliability. Therefore, the design is suitable for banking environments where continuous network availability and data reliability are critical.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

ANSWER:

Problem Analysis

A small classroom computer lab requires a **low-cost and simple network** for basic activities such as Internet browsing, file sharing, and accessing educational resources. Since the budget is limited and network traffic is relatively low, an expensive network infrastructure is unnecessary.

Solution Design

A **bus topology** is suitable for a small classroom lab because all computers share a common communication backbone. This reduces the amount of networking hardware and cabling required.

In a traditional bus network, a **50-ohm RG-58 coaxial cable** acts as the main backbone. Each computer is connected to the backbone using a **T-connector and BNC connector**. **50-ohm terminators** are placed at both ends of the cable to prevent signal reflections.

For a modern low-cost implementation, an **8-port or 16-port unmanaged Ethernet switch** with Cat5e/Cat6 cables can be used. Although this is technically a star topology, it provides a similar simple and inexpensive network arrangement for a classroom.

Basic structure:

Computer 1 — Computer 2 — Computer 3 — Computer 4 — Computer 5 — Computer 6

All computers share the common network communication path.

Technical Implementation

For the traditional bus implementation:

- **RG-58 coaxial cable** is used as the backbone.
- **T-connectors** connect computers to the backbone.
- **BNC connectors** provide the physical cable connections.
- **50-ohm terminators** are installed at both ends of the backbone.
- Network interface cards with appropriate coaxial support are required.

For a modern implementation, Cat5e/Cat6 UTP cables and an inexpensive unmanaged Ethernet switch can be used.

No dedicated server is required. A teacher's computer can provide shared folders and resources using **Windows or Linux file-sharing features**.

Use of Tools and Technologies

The following technologies can be used:

- **Bus topology** – provides a simple shared communication medium.
- **RG-58 coaxial cable** – traditional bus network backbone.
- **T-connectors and BNC connectors** – connect computers to the backbone.
- **50-ohm terminators** – reduce signal reflection.
- **Cat5e/Cat6 cables** – modern Ethernet connectivity.

- **Unmanaged switch** – provides an inexpensive modern alternative.
- **Windows/Linux file sharing** – allows computers to share files and resources.

Problem-Solving Approach

The main challenge is creating a functional network while keeping the cost low. Bus topology addresses this by reducing the requirement for individual connections to a central networking device.

It is appropriate for a supervised classroom because network usage is generally light. Students can perform basic browsing, file sharing, and educational activities without requiring an advanced network infrastructure.

Innovation and Scalability

The network can be upgraded when the classroom grows. A modern Ethernet switch can replace the traditional coaxial bus, and additional switches or wireless access points can be introduced when required.

The teacher's PC can also be configured as a shared-resource computer, reducing the need for a dedicated server and lowering the overall implementation cost.

Testing and Validation

The network should be tested using:

1. **Ping test** – verify communication between computers.
2. **File-sharing test** – verify that students can access shared files.
3. **Internet connectivity test** – confirm Internet access.
4. **Cable inspection** – check for damaged or loose connections.
5. **Backbone test** – verify that the bus cable and terminators are correctly connected.
6. **Failure test** – identify the effect of a cable failure on connected computers.

Documentation and Presentation

The network diagram should show all classroom computers, the common backbone cable, connectors, and terminators. The documentation should include cable types, network configuration, IP addresses, and testing results.

For a modern implementation, the diagram should show the unmanaged switch and the Cat5e/Cat6 connections.

Teamwork and Collaboration

The work can be divided among team members. One member can install and check the cables, another can configure the computers and IP addresses, another can test file sharing and Internet connectivity, and another can prepare the network documentation.

Advantages

- Low implementation cost
- Simple network design
- Requires less networking equipment

- Suitable for small networks
- Easy to set up for basic classroom activities
- No dedicated server is required

Limitations

- A backbone cable failure can affect the entire traditional bus network.
- Shared communication can lead to collisions in traditional bus Ethernet.
- Network performance decreases as more devices are added.
- Troubleshooting a physical bus can be more difficult.
- It has limited scalability compared with modern switched networks.

Conclusion

A **bus topology** is a suitable low-cost solution for a small classroom lab with light network usage. Its simple shared-medium design reduces hardware and cabling costs. However, because traditional bus networks are less reliable and scalable, a modern inexpensive Ethernet switch is a more practical alternative for an actual classroom deployment.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

ANSWER:

Problem Analysis

A large university campus contains thousands of users distributed across **lecture halls, laboratories, libraries, hostels, administrative offices, and other buildings**. A single network topology cannot efficiently satisfy all requirements. The network must provide high speed, reliability, security, easy management, and scalability.

Solution Design

A **hybrid topology** is the most suitable solution because it combines different topologies according to the requirements of each part of the campus.

The proposed design uses:

- **Ring/partial mesh topology** at the campus backbone for redundancy.

- **Star topology** within individual buildings for easy management.
- **Tree/hierarchical topology** for connecting different floors and departments.
- **Bus-like or simple switched structures** where appropriate in small laboratories.

Network structure:

Internet → Core Routers → Core Switches → Building Distribution Switches → Access Switches → Computers / Printers / Wi-Fi APs

The campus backbone uses high-speed fiber connections between core routers and multilayer switches. Each building has a distribution switch with **dual fiber uplinks** to the campus core.

Technical Implementation

At the campus core, **single-mode fiber optic cables** can be used for long-distance connections between buildings. High-speed links such as **10 Gbps or 40 Gbps** can provide sufficient bandwidth for large amounts of campus traffic.

Each building has a **distribution switch** connected to the campus core through redundant fiber links. Floor-level access switches connect to the distribution switch, and individual computers, printers, and other devices connect to these access switches using **Cat6 UTP cables**.

Wireless access points are connected to access switches using **Power over Ethernet (PoE)**. PoE allows both data and electrical power to be delivered through the same Ethernet cable.

The design follows a hierarchical **three-layer architecture**:

1. **Core Layer** – High-speed campus backbone and routing.
2. **Distribution Layer** – Connects buildings and provides traffic control.
3. **Access Layer** – Connects end-user devices.

Use of Tools and Technologies

The following technologies are used:

- **Hybrid topology** – combines multiple topologies according to requirements.
- **Fiber optic cables** – provide high-speed backbone connectivity.
- **Multilayer switches** – perform switching and routing.
- **Core routers** – provide connectivity between campus networks and external networks.
- **Cat6 UTP cables** – connect end devices.
- **PoE** – powers wireless access points through Ethernet cables.
- **Wireless Access Points** – provide campus-wide Wi-Fi connectivity.
- **VLANs** – can separate students, faculty, administration, guests, and laboratory networks.

Problem-Solving Approach

The major challenge is supporting thousands of users while maintaining network reliability and performance. The hybrid design solves this by assigning different topologies to different network levels.

If one fiber link fails, the redundant backbone path can continue carrying traffic. If a single computer or access cable fails, only that device is affected. This provides **fault isolation** at the building and user levels while maintaining redundancy at the campus core.

Innovation and Scalability

The design is highly scalable. When a new building is constructed, a new distribution switch can be installed and connected to the campus core using fiber. Existing buildings do not need to be redesigned.

Additional access switches and wireless access points can also be added as the number of students and staff increases.

PoE reduces installation complexity because wireless access points do not require separate electrical power cables.

Testing and Validation

The campus network should be tested using:

1. **Ping tests** – verify connectivity between buildings.
2. **Traceroute tests** – verify routing paths through the campus network.
3. **Fiber link testing** – check the performance and integrity of backbone links.
4. **Redundancy testing** – disconnect one fiber link and verify that traffic uses the backup path.
5. **Wi-Fi testing** – check wireless coverage in classrooms, libraries, hostels, and common areas.
6. **Bandwidth testing** – measure network performance during high traffic.
7. **PoE testing** – verify that wireless access points receive both power and network connectivity.

Documentation and Presentation

A detailed network diagram should show the **campus core, routers, multilayer switches, building distribution switches, access switches, fiber links, Wi-Fi access points, and end devices**.

Documentation should include IP addressing, VLAN assignments, cable types, switch locations, redundancy paths, and testing results. Proper documentation makes future troubleshooting and network expansion easier.

Teamwork and Collaboration

The implementation can be divided among different teams. One team can design the campus backbone, another can configure building distribution and access switches, another can install and configure wireless access points, and another can perform testing, security configuration, and documentation.

Advantages

- Highly scalable for large campuses
- Provides backbone redundancy
- Easy management within buildings
- Supports thousands of users
- High-speed fiber connectivity
- Easy addition of new buildings

- Supports wired and wireless devices
- Provides better fault isolation
- PoE reduces cabling requirements

Limitations

- Higher initial installation cost
- Requires skilled network administrators
- Fiber installation can be expensive
- Network configuration is more complex than a single topology
- Regular maintenance and monitoring are required

Conclusion

A **hybrid topology** is the best choice for a large university campus because it combines the advantages of different network topologies. A **fiber-based ring or partial mesh** provides redundancy at the campus backbone, while **star and hierarchical structures** provide simple management within buildings. This design offers **high performance, reliability, scalability, and flexibility**, making it suitable for a modern university campus.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

ANSWER:

Problem Analysis

A company with departments such as HR, Finance, Sales, and IT requires a network that provides efficient communication while maintaining security, performance, reliability, and scalability. Each department may have different users and security requirements, so the network should logically separate departmental traffic.

Solution Design

A hierarchical network architecture using Layer 2 access switches and a Layer 3 core switch or router is suitable.

Each department is connected to a dedicated Layer 2 access switch. These switches connect to a central Layer 3 switch, which performs inter-VLAN routing and enables controlled communication between departments.

Network structure:

Internet → Router/Firewall → Layer 3 Core Switch → Layer 2 Access Switches →

- HR Department
- Finance Department
- Sales Department
- IT Department

Each department is assigned a separate VLAN to logically isolate its network.

VLAN Design

Department	VLAN	Example Subnet
HR	**VLAN 10**	**192.168.10.0/24**
Finance	**VLAN 20**	**192.168.20.0/24**
Sales	**VLAN 30**	**192.168.30.0/24**
IT	**VLAN 40**	**192.168.40.0/24**

This separation reduces unnecessary broadcast traffic and improves security.

Technical Implementation

Layer 2 switches are installed at the access layer to connect computers, printers, and other devices within each department.

A Layer 3 switch is placed at the core layer and configured for inter-VLAN routing. This allows authorized communication between departments.

For example, the Finance VLAN can be prevented from accessing sensitive HR resources unless permission is explicitly configured.

A router/firewall connects the internal network to the Internet and protects the company network from unauthorized external access.

Proper IP addressing and subnetting are used to efficiently allocate addresses and simplify network management.

Use of Tools and Technologies

The network uses:

- Layer 2 switches – connect departmental devices.
- Layer 3 switch – performs routing between VLANs.
- VLANs – logically separate departments.
- Router – connects the company network to external networks.
- Firewall – protects the internal network.
- IP addressing and subnetting – organize network addresses efficiently.
- Redundant links – provide backup connectivity.
- DHCP – can automatically assign IP addresses to client devices.

Problem-Solving Approach

The main challenge is allowing departments to communicate when required while preventing unnecessary or unauthorized access.

VLANs solve this problem by separating departmental traffic. Inter-VLAN routing on the Layer 3 switch allows communication only when appropriate routing and security rules are configured.

For example, employees in Sales can communicate within their VLAN, while access to Finance resources can be restricted using ACLs (Access Control Lists).

Innovation and Scalability

The hierarchical design makes the network easy to expand. If the company adds a new department, a new VLAN and access switch can be configured without redesigning the entire network.

Redundant links can also be added between access switches and the core switch to improve reliability. Network monitoring tools can be used to identify congestion, failures, and unusual traffic.

Testing and Validation

The network should be tested using:

1. **Ping test –** verify communication within each VLAN.
2. **Inter-VLAN connectivity test –** verify authorized communication between departments.
3. **VLAN isolation test –** confirm that unauthorized departmental access is blocked.
4. **IP configuration test –** verify correct IP addresses, subnet masks, and gateways.
5. **Internet connectivity test –** confirm access through the router/firewall.
6. **Redundancy test –** disconnect a backup link and verify continued connectivity.
7. **Security testing –** verify ACL and firewall rules.

Documentation and Presentation

The network diagram should clearly show the Internet, router/firewall, Layer 3 core switch, departmental Layer 2 switches, VLANs, and end devices.

Documentation should include VLAN numbers, IP addresses, subnet masks, gateway addresses, switch connections, routing configuration, firewall rules, and testing results.

Teamwork and Collaboration

The implementation can be divided among team members. One member can configure VLANs, another can configure the Layer 3 switch and inter-VLAN routing, another can configure the router/firewall, and another can perform testing and documentation.

Advantages

- Improved network security
- Reduced broadcast traffic
- Easy departmental management
- Efficient communication
- Easy troubleshooting
- Supports network expansion
- Provides better performance
- Allows implementation of access-control policies
- Supports redundancy

Limitations

- Requires skilled configuration
- Layer 3 switches and managed switches increase initial cost

- Incorrect VLAN or routing configuration can cause connectivity problems
- Regular monitoring and maintenance are required

Conclusion

A hierarchical network using Layer 2 access switches, a Layer 3 core switch, VLANs, routing, and a firewall is an effective solution for a company with multiple departments. It provides security, efficient communication, scalability, reliability, and centralized management, while allowing the company to expand its network easily in the future.